Image Denoising by Anisotropic Diffusion with Inter-scale Information Fusion[1]

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Abstract— Anisotropic partial differential equations (PDEs) based schemes for denoising digital images are fast becoming an indispensable tool in computer vision problems. In this paper we propose to denoise noisy images via such multiscale anisotropic diffusion. In general, digital images contain objects of multiple scales and denoising them without destroying edges is one of the main objective in early computer vision problems. Unlike the previous approaches, which discard the multiple scale based images produced by anisotropic PDE, we utilize information contained in them. By effectively combining the inter-scale details, the proposed scheme improves upon the noise removal and detail preservation properties over other schemes. Numerical results indicate that the scheme achieves good denoising with edge preservation on a variety of images.

Keywords: anisotropic diffusion, image restoration, inter-scale, information fusion, edge maps, Gaussian noise

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1. INTRODUCTION

In image restoration the primary task is to remove noise without changing the multi-scale nature of digital images. There exists various methods for image denoising and variational-partial differential equation (PDE) based schemes [1] are powerful in this regard. Perona and Malik (PM) [2] proposed the following anisotropic diffusion PDE,

$$\begin{cases} \dfrac{\partial u(x,t)}{\partial t} = \operatorname{div}(g(|\nabla u(x,t)|)\nabla u(x,t)) \\ u(x,0) = u_0(x), \end{cases} \tag{1}$$

where $u_0 : \Omega \subset \mathbb{R}^2 \to \mathbb{R}$ is the noisy input image and $g : \mathbb{R} \to [0,\infty)$ is a nonlinear function which controls the diffusion within flat regions. The artificial time parameter t is akin to the diffusion evolution time from physical systems and the evolution of the initial image $u_0(x)$ towards a final $u(x,T)$, $T > 0$ achieves denoising using the diffusion function $g(|\nabla u|)$ as a guidance. Good numerical results are reported in [2] and its variants and other aspects were studied in recent years [3–6], see [1] for a review. The PDE in Eq. (1) gives rise to what is known as a nonlinear scale-space $\{u(x,t)\}_{t=1}^{\infty}$ of the input image $u_0(x)$ and the parameter t is related to scales of objects present and is known as the scale parameter. Typically the PDE based schemes are stopped at a time T and the image $u(x,T)$ is considered the de-noised version. Due to the multi-scale nature of images, there exists objects of different scales and capturing them in a single scale gives unsatisfactory results.

In this paper we concentrate on anisotropic PDE based image denoising (1) and study how we can improve it further using fine to coarse smoothed images obtained as part of nonlinear PDE evolution. We propose to make use of salient edges information present in various scale images by integrating them in the final step. This is done by fusing the information provided by the fine to coarse scale images obtained from the PDE (1) into the diffusion coefficient. This updated diffusion coefficient is utilized once in a few iterations and experimental results illustrate we obtain better results in terms of edge preservation and signal to noise ratio improvement, see Fig. 1 for an example. Experimental results on noisy images are given to highlight the improvement obtained by utilizing inter-scale information fusion compared with traditional gradient approaches.

The rest of the paper is organized as follows. Section 2 introduces the PDE based denoising scheme. Section 3 presents the denoising results which shows the advantages the scheme on noisy digital images. Finally, Section 4 gives the conclusions indicating possible future directions.

2. DIFFUSION MODEL WITH INTER-SCALE FUSION

2.1. Gradient as Edge Map

The diffusion function $g(|\nabla u|)$ in the PM PDE (1) uses the information provided by magnitude of the

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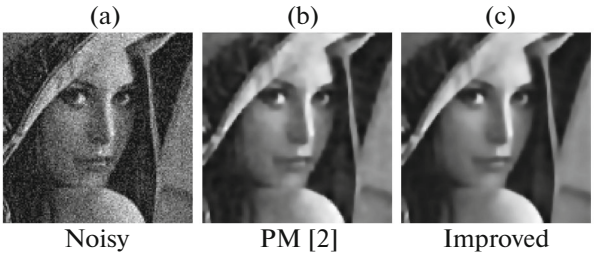

Fig. 1. Diffusion denoising improves by utilizing inter-scale information fusion. (a) Noisy *Lena* 128×128 image consisting synthetic and real sub images (Gaussian noise σ_n) (b) Perona and Malik scheme (1) result with $g(s) = (1 + (s/30)^2)^{-1}$, after 50 iterations (c) Our inter-scale master edge map based result, after 50 iterations. Notice the improvement in terms of staircasing alleviation in flat regions.

gradient to lower the diffusion (smoothing) near edges in Eq. (1). Figure 2 shows the effect of this when we implement the PM scheme on an image created by combining synthetic constant, ramp, and real sub-images. Original *Montage* image shown in Fig. 2a, it is corrupted with additive Gaussian noise of $\sigma = 20$ and is given in Fig. 2b. PM equation (1) with only the diffusion coefficient g drives the process, removes noise and preserves edges (Fig. 2c) but induces false edges in constant, ramp regions, this effect is known as *staircasing* and can be detrimental in further image processing tasks. The magnitude of the gradient image (Fig. 2c left, where whiter pixels represent maximal value of $|\nabla u|$) shows that it is not reliable under noise and loss of spatial coherence such as edge connectivity and continuity is high. Further iterations of anisotropic diffusion either amplify noise in texture areas or over-smooth small scale edges. Thus, designing an efficient diffusion coefficient which not only depends on noisy gradient measurements is important in obtaining better restorations.

2.2. Inter-scale Information

We propose to use the following steps to improve the anisotropic diffusion scheme using inter-scale information through the scale space. More precisely,

1. Starting with noisy image $u(x, 0) = u_0(x)$ run the PDE (1) for a few iterations, say T_1 and obtain $\{u(x,t)\}_{t=1}^{T_1}$.

2. Next, compute gradient images $\{\nabla u(x,t)\}_{t=1}^{T_1}$ and sum them across scales to obtain a master gradient map

$$\tilde{\nabla} u(x, T_1) := \cup_{t=1}^{T_1} \nabla u(x,t). \tag{2}$$

3. At iteration $T_1 + 1$ use the master gradient map $\tilde{\nabla} u(x, T_1)$ in the diffusion coefficient $g\left(|\tilde{\nabla} u(x, T_1)|\right)$ to run the PDE (1).

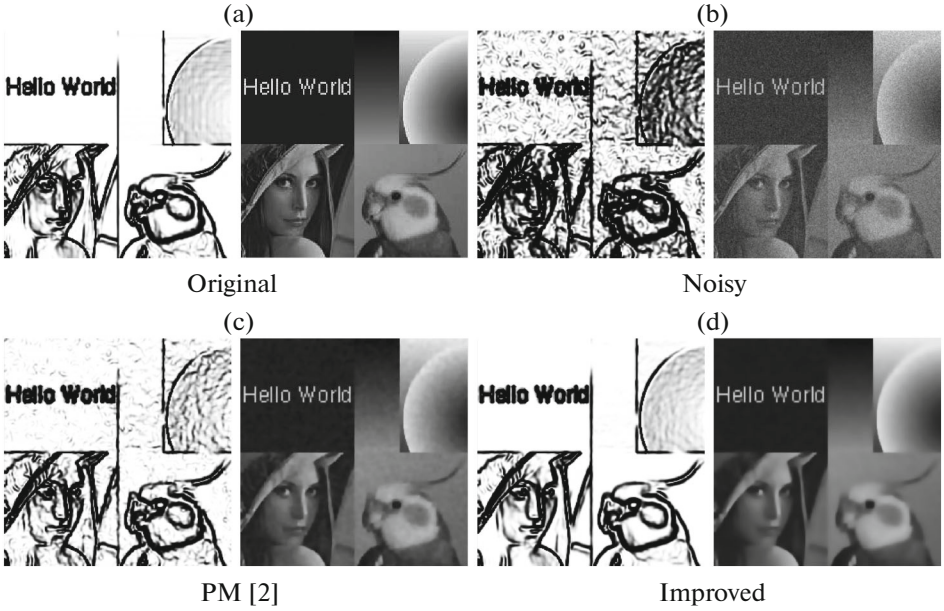

Fig. 2. Better restorations can be obtained by using improved edge maps in the diffusion process. (a) Gradient map of the original image $|\nabla u_O|$ (b) Noisy *Montage* 256×256 image (c) Gradient magnitude $|\nabla u_{49}|$ used in PM equation at the iteration tP (d) Restoration result using PM equation (1) with $g(s) = (1 + (s/10)^2)^{-1}$, after 50 iterations (e) Inter-scale details fused edge map $\tilde{\nabla} u(x, 40)$ (f) Our scheme result with $g(s) = (1 + (s/30)^2)^{-1}$, after 50 iterations.

4. Repeat Steps 1-3 starting with initial image $u(x,T_1)$ until iteration time T_2.

5. Continue computing the master gradient map $\tilde{\nabla} u$ at particular iterations $T_1 \le T_2 \le \ldots$.

6. Stop when it reaches the set maximum iteration T.

Note that the master gradient map (3) is computed only at a chosen time steps and it reverts back to the original diffusion coefficient if done at every iteration, i.e., if $T_1 = 1, T_2 = 2, \ldots, = T$.

Figure 2(d) shows the result of improved inter-scale edge information fusion. As can be seen, the salient edges are retained and the master edge map (Fig. 2d, left) show no staircasing effects. A closer look at the ramp regions is given in Fig. 3. Compared to the result of PM PDE (1) with traditional gradient map (Fig. 3b), our inter-scale information fusion improvement (Fig. 3c) obtains a staircasing free result. This can further be seen in the 1D line profile taken across the ramp region (Fig. 3d), where the improved scheme removes spurious noise without creating artifacts and obtains best result visually. This is further evidenced quantitatively by Table 1 where various error metrics are given for this experiment. Note that these error metrics are computed using noise-free (original, u_O) synthetic *Montage* image.

3. EXPERIMENTAL RESULTS

All the images are normalized to the range [0, 1]. We consider the standard finite difference implementation differential equations with adiabatic boundary conditions using the explicit Euler scheme [7]. Computations are done on a desktop computer with Pentium IV, 2.45 GHz processor. It took less than 0.5 seconds for 100 iterations of our scheme due to inter-scale information fusion being applied at only few levels.

The following two schemes are used for comparisons:

• Perona and Malik scheme (PM) [2]

$$\frac{\partial u(x,t)}{\partial t} = \operatorname{div}\left(\frac{\nabla u(x,t)}{1+|\nabla u(x,t)|^2/K^2}\right) \tag{3}$$

• Rudin, Osher, and Fatemi's total variation scheme (TV) [8] (with regularization)

$$\frac{\partial u(x,t)}{\partial t} = \operatorname{div}\left(\frac{\nabla u(x,t)}{\sqrt{\varepsilon^2+|\nabla u(x,t)|^2}}\right) \tag{4}$$

The parameter $K = 10$ is fixed in Eq. (4) and TV scheme regularization parameter $\varepsilon = 10^{-4}$ is set in Eq. (5) in all the experimental results reported here. The maximum iteration (T) is set at 50 for the PM PDE (1) and 100 for the TV scheme for all the results reported here. We use $T_1 = T_2 = \ldots 10$, that is, we use the master gradient map (3) only at every 10 iterations. We use the following error metrics to quantify the

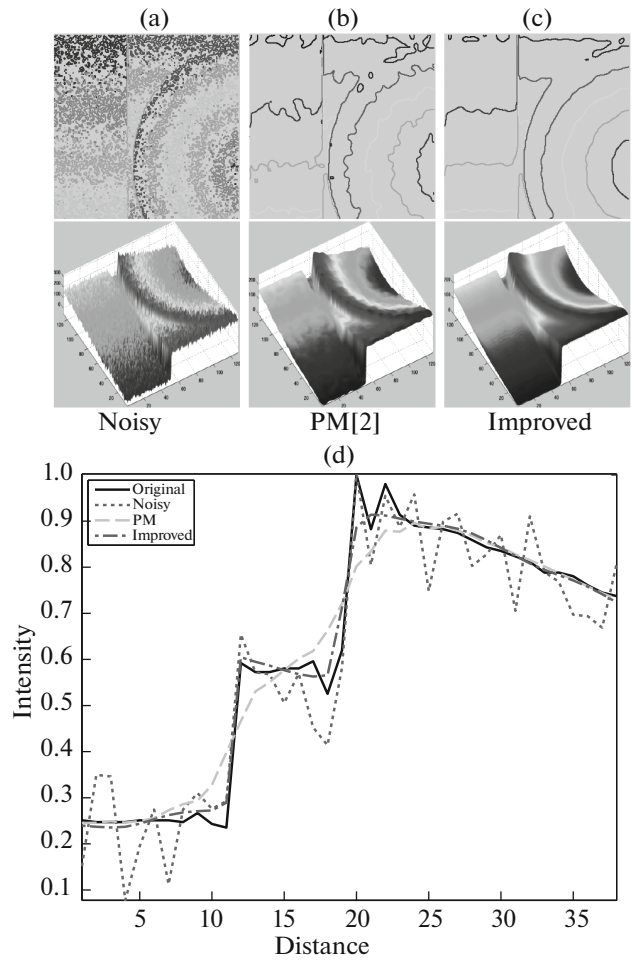

Fig. 3. Close up on the *Montage* image highlighting the ramp regions from Fig. 2. (a) Noisy (b) PM [2] result (c) Improved PM with inter-scale information fusion (d) A line taken across the ramp region shows clearly that our improved version does not over-smooth the edge regions and devoid of staircasing artifacts.

Table 1. Various error metrics for the restoration of noisy *Montage* 256×256 image in Fig. 2 using the PM [2] PDE and improved version based on inter-scale edge fusion method proposed in Section 2

Metric	Noisy	PM [2]	Improved
ISNR	0.0000	1.6569	8.1678
SNR	14.756	16.413	22.924
PSNR	22.084	23.740	30.251
MSSIM	0.3561	0.8223	0.8915
MSE	402.454	274.81	61.367
RMSE	20.0613	16.577	7.8337
MAE	16.0012	7.9500	4.8185
MAX	84.3010	176.44	138.90

Table 2. Comparison of error metrics for various test images. Results on noisy (Gaussian noise $\sigma_n = 20$) images using the PM [2] and TV [8] PDEs and their improved versions based on inter-scale edge fusion method proposed in Section 2.2.2. Best values are given in boldface

			Previous		Improved	
Image	Metrics	Noisy	PM [2]	TV [8]	PM	TV
Lena	PSNR	22.08	28.19	28.71	28.88	**30.09**
	SSIM	0.4614	0.8283	0.8346	0.8492	**0.8615**
	MAE	16.001	6.4903	6.4655	5.5373	**5.1215**
Barbara	PSNR	22.10	23.05	24.33	23.16	**24.56**
	SSIM	0.7550	0.7934	0.8344	0.8154	**0.8490**
	MAE	15.988	11.098	9.8199	10.486	**9.0095**
C.man	PSNR	22.08	26.84	27.85	27.29	**28.74**
	SSIM	0.3861	0.8010	0.8179	0.8341	**0.8493**
	MAE	16.001	6.7621	6.295	5.6865	**4.9136**
Peppers	PSNR	22.08	29.39	29.81	30.27	**31.62**
	SSIM	0.4169	0.8671	0.8585	0.8941	**0.8921**
	MAE	16.001	5.8470	5.8803	4.8295	**4.4325**
House	PSNR	22.08	31.14	31.10	32.40	**33.41**
	SSIM	0.3372	0.8533	0.8467	0.8860	**0.8847**
	MAE	16.001	4.8903	4.9700	3.7515	3.3896
Boat	PSNR	22.10	28.59	29.04	29.13	**30.00**
	SSIM	0.7362	0.8781	0.8820	0.9028	**0.8920**
	MAE	15.988	6.3820	6.2715	5.6222	**5.2710**

improvement of our inter-scale fusion method when compared with PM method [2] and TV [8]:

• Improved signal to noise ratio (ISNR), signal to noise ratio (SNR), peak signal to noise ratio (PSNR), and structural similarity measure (SSIM) [9] with higher values indicating better performance.

• Mean squared error (MSE), root mean squared error (RMSE), mean absolute error (MAE), and maximum absolute difference (MAX) with lower values indicating better performance.

Figure 4 provides comparison of our improvement on various noisy images, which are obtained by adding Gaussian noise σ_n, with PM PDE (Eq. (4)) along with ISNR values. Similar improvements were observed with TV PDE (Eq. (5)) and omitted here for brevity. In Table 2 we provide SSIM, PSNR (dB), and MAE values for various images and compare with traditional gradient driven diffusion schemes. Inter-scale edge maps fusion improves the restoration results and for both PM [2] and TV [8] we obtain better results. Our approach is general in the sense that we can incorporate inter-scale gradient fusion approach to other anisotropic diffusion models.

One of the limitations of the improved scheme is the lack of texture preservation, see for example the part of feather, hair in Fig. 2d, and *Barbara* result in Fig. 4 (first row). Better diffusion coefficient modeling can improve our approach [10] and a separate indicator function is required to handle partially textured images and is left as a future work.

(0.9494 → 1.067)

(4.7542 → 5.2042)

(7.3018 → 8.1831)

(9.0564 → 10.3193)

(6.4927 → 0.9760)

Fig. 4. Denoising results improve with inter-scale information fusion. (Left) PM scheme [2], see Eq. (3) (Right) Improved using our inter scale information fusion method. ISNR values with respect to noisy images are given at each result. Note the staircasing artifact reduction in flat regions along with edge preservation with the improved schemes.

4. CONCLUSIONS

In this paper, we proposed a inter scale fusion based image denoising scheme utilizing anisotropic diffusion. We combine information provided by multiple scale images into a time dependent PDE and obtain better denoising than a single scale based diffusion schemes. Fine to coarse scale images are progressively fused together to give a clear edge map which in turn gives good denoising results. Extending the proposed inter-scale information fusion based anisotropic diffusion to handle mixed noise [11] and color images [12] are interesting directions which needs to be explored further.

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